

run for 70 to 78 hours with a single charge, has a processor block and analogue front-end block, and 1.5kb of SRAM. Some devices also include EEG bands, heart rate monitor, atrial fabrication detector, and more.

Reducing dynamic power

There are too many power dissipating components in an integrated system. One of the approaches to minimise power consumption is to reduce the dynamic power. Dynamic power is the power consumed by the circuit when it is switched on. It comprises switching power and short-circuit power.

The short-circuit power is related to current and flows directly from supply voltage to the ground. When both NMOS and PMOS networks are momentarily switched on, the loss of short-circuit power is relatively small and switching power dominates the dynamic power consumption. Here, the switching power is kept in a quadratic manner with the voltage supply.

Quadratic manner relates to the comparative behaviour of voltage, total current, leakage current, and dynamic power when a graph of energy with respect to the voltage is plotted.

To further reduce the power consumption, frequency scaling can be done in conjunction with voltage scaling. Reducing voltage and frequency is referred to as dynamic voltage frequency scaling. It significantly reduces the dynamic power according to a device's required performance.

However, increasing the voltage frequency scaling and operating it at a sub-threshold voltage increases the leakage energy (since the device operates longer and slower in this region). Therefore, it is better to operate near a sub-threshold region where both leakage and dynamic power are optimal for a highly power-constraint wearable device.

Voltage scaling

Supply voltage scaling is done to produce the desired voltage level for each block. Since these blocks consist of an SRAM that operates at a minimal voltage of 0.9V in 45nm CMOS technology, there is a requirement

for power management (with respect to the blocks). Whereas digital blocks can be operated at 0.4V to minimise power consumption when data is transferred via a wireless transmitter-receiver. Thus, multiple voltage levels are required in a single chip. So, the block requires a voltage level of may be up to 1V.

This type of voltage scaling can give the required power to the blocks. Otherwise, if a block that is not supposed to consume too much power gets a large voltage or current, the battery will not last long.

There are some good architectures out there like the 16-core ARM micro-processor where each core requires a dynamic voltage frequency scaling (DVFS) block. Voltage regulators that convert the input voltage to the desired output voltage level can be used. Integration of multiple voltage regulators is needed to support dynamic voltage scaling.

Thus, the area of the power management unit is dominated by voltage regulators. As the number of desired regulated voltage levels increases, the number of voltage regulators also increases, which results in a large area overhead. Therefore, supporting multiple regulated voltage levels with a minimum number of voltage regulators needs to be explored.

Leakage power

Leakage power is the power consumed by the circuit when it is inactive. Although some activities can be performed in the sleep mode, these are limited to the microcontroller only. With soft programming you can control the sensors and the transmitter block, that is, when it has to power up and when it has to go into deep sleep mode. That will save a lot of power.

The leakage power can be determined when the circuit is operating but a part of it is inactive. It means that when the IR sensor, pulse rate sensor, and SpO2 sensor try to send data to the microcontroller, it is not received properly because some part of the controller is still inactive. This type of power is also called Active Leakage, that is, some blocks are active while some are inactive.

Due to the large number of

transistors in a single chip, leakage power becomes critical. This becomes important especially when there is a multi-core SoC, that is, an SoC that has cross-voltage and performs some virtual circuit switching operations (data transfer from one place to another with the help of low-latency as well as low-power consumption).

Although CMOS transistor size is shrinking and the device speed is improving, the leakage current is also rapidly increasing. This is due to the channel lengths in CMOS and PMOS devices. The impact of leakage power becomes worse as the average power consumption is dominated by power during the adder time. Therefore, duty cycle technique is employed to ensure the system enters a low power mode during idle time when the supply voltage is reduced for some blocks and supply discontinued for those blocks that do not require it.

This arrangement with the help of soft programming saves lots of power while performing operations like walking and trekking as well as sending data via Bluetooth LE to mobile based applications. When the device wakes up momentarily, it consumes power in milliamps to perform repetitive tasks of sensing, processing, and communicating, and then falls asleep till the next task is assigned.

The average power consumption of a device depends on its p-active state when the device is in its active mode. The average power consumed by a device thus depends on the total of the active power + power consumed by the transceiver + power consumed by features that are always on (even in sleep mode).

To manage power effectively, you have to focus on all these elements. Focus on every block that is a part of the device like sensors for processing blocks, transmitting-receiving blocks, and voltage regulator blocks, and assemble them in such a way that the device consumes minimum possible power. **EFY**

The article is based on the talk 'Balancing Power & Conservation In Smart Wearables' by Abhijeet Bhagat, CEO, Atmen Technovention, which was presented during January edition of Tech World Congress 2021.